

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

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Abstract

How could activity distributed across cortex support one experienced situation? We separate three requirements: *coherence* (collective phase order), *compatibility* (local descriptions agree where their domains overlap) and *self-representation* (the state carries an accurate reconstruction of itself). Coherence supplies neither of the others, and five results establish that with no cortical hypothesis. Synchrony bounds content disagreement only through encoders shared on overlaps, and then only at the patch the bound is read on, not over the whole population. The resultant does not determine the state: a winding field and a uniform one are stationary at one coupling, at opposite ends of that observable. A unique fixed point is not a self-model, since a contraction can discard its input: accuracy across a family of states requires distinguishable codes. Where the encoding is a region's restriction, that resolution is demanded of the region before any readout is chosen. And agreement about a change takes communication. Four of the five are checkable for a machine candidate, which fixes what a claim of machine unity or self-representation commits its maker to measuring. Under stated hypotheses, compatible

local states then glue to a unique global state that an accurate contracting reconstruction fixes uniquely. Installed energy bounds the mean coupling a declared actuator supplies, so a coherent stationary branch requires it to exceed $2D/\kappa$; κ is derived here for neither cortex nor a processor. A Lean 4 development carries these results and their composition, with the eight connecting hypotheses as explicit arguments, no incomplete proofs, and an audited axiom footprint no wider than its libraries'. Further controls bound what observation can decide: distinct mechanisms share stationary phase distributions, and coherent adaptation can lose the structure it was meant to acquire. One prediction needs no absolute calibration: onset delay scales as the square root of the rate at which coupling crosses threshold, which a within-subject emergence-from-anaesthesia design can test; the finite-size estimate here is biased below $1/2$ by its escape criterion, as a deterministic reference confirms. We propose the glued state and its internal reconstruction as correlates of unity and a minimal self, and an extracellular-field hypothesis linking awakening to rising coupling. Testing it requires calibrated geometry, resolved phase dynamics and validated decoders.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Successful coordination would still leave a representational question: what makes the coordinated activity a description of one situation, with a perspective on that situation?

We distinguish three requirements. *Coherence* concerns collective phase

11 order. *Compatibility* concerns whether local descriptions agree where their
12 domains overlap. *Self-representation* concerns whether the resulting state
13 carries an accurate internal reconstruction of itself. We propose the latter
14 two relations as formal correlates of conscious unity and a minimal self; their
15 interpretation requires evidence beyond the phase model.

16 Unity has been approached differently elsewhere: integrated information
17 theory treats integration and exclusion as constitutive (Tononi et al., 2016),
18 global-workspace accounts tie availability to broadcast (Baars, 1997; Dehaene
19 et al., 2014), and predictive-processing accounts derive both from a hierar-
20 chical generative model (Friston, 2010). Here the three requirements are
21 kept apart rather than derived from one another, so that each carries its own
22 assumptions and admits its own measurement; Section 9.4 returns to the
23 comparison.

24 Consider reaching for a cup. A visual description of its location and a
25 description of the hand’s relation to it can agree about where the cup is while
26 contributing different contents. Synchronizing the two processes would leave
27 disagreement about that shared quantity unresolved. An internal representa-
28 tion must additionally locate the perceiver within the represented situation.
29 This example motivates the requirements; the mathematical constructions
30 use spatial measures and fields in real normed vector spaces, whose identifi-
31 cation with perceptual content is a further modelling task.

32 The mathematical contribution is to specify connections between these
33 requirements and the assumptions each connection needs. A declared ac-

34 tuator’s stored energy bounds its mean coupling. Phase encoders shared on
35 overlaps convert phase order into a bound on content disagreement. Compat-
36 ible spatial measures have a unique global extension; approximate agreement
37 controls a specified global selection. A contracting reconstruction makes that
38 extension its unique fixed point only if it accurately reconstructs the exten-
39 sion. Accuracy across a family of states imposes a further condition on the
40 information retained by the encoding.

41 The empirical contribution is to identify what observations can decide.
42 Stationary phase distributions can be shared by stronger coupling, weaker
43 diffusion and a matched common drive. Finite ramps change the observable
44 onset. A winding field is stationary at the same coupling as a uniform one and
45 has the opposite resultant, so that observable does not determine the state.
46 Coherent adaptation can lose alignment with a designated environmental
47 template, and a compatibility estimator needs diagnostics that distinguish
48 content from decoder artifacts. These findings determine the measurements
49 needed to test the proposed connections.

50 Three of the ingredients are standard: the synchronization threshold
51 $K_c = 2D$ for the noisy mean-field model (Sakaguchi, 1988), the unique gluing
52 of compatible local sections, and the Banach fixed-point theorem (Banach,
53 1922). Four things here are not. The installed-energy condition bounds
54 mean coupling by the stored energy of a declared actuator, which makes a
55 resource condition necessary for coherence on the stationary branch. The
56 coherence-to-content bound converts phase order into a statement about de-

57 descriptions rather than about timing, and its patch-local form is what keeps
58 it from being vacuous in a large population. A stability criterion on the
59 reweighted coupling kernel, together with an existence condition on an am-
60 plitude layer, settles when a low resultant reports weak coupling and when
61 it reports a winding field. And the composition is machine-checked together
62 with a witness discharging all eight hypotheses at once, which is what makes
63 their conjunction non-empty rather than a list no object satisfies.

64 Granting those premises, collected in Table 1, the account runs as follows.
65 Extracellular contraction on awakening raises endogenous field amplitude
66 and with it the effective coupling between cortical oscillators. Once that
67 coupling passes $2D$, a coherent phase branch exists, and from an initial spread
68 within a quarter turn the population settles onto it; other initial data leave
69 windings stationary at the same coupling. Encoders shared where cortical
70 territories overlap turn that phase order into agreement about the quantities
71 the territories describe in common, so the local descriptions glue: there is
72 one state of which each is a restriction. Part of that state encodes the state
73 itself, and a readout of that part reconstructs it accurately enough, and
74 sensitively enough across the states that matter, to make it the unique state
75 the system models itself as being. On this account the glued state is what
76 makes distributed activity one situation, and its accurate self-reconstruction
77 is what gives that situation a perspective. Every step of it is conditional.

78 Section 2 collects the results that hold whatever physical field is proposed
79 and whatever substrate carries it; a reader who rejects the cortical hypothesis

80 of Section 8 still has to accommodate them. Sections 3–5 then develop the
 81 three requirements together (Fig. 1). Section 6 composes them, stating the
 82 thermodynamic constraints and all eight connecting hypotheses per branch;
 83 its supporting models of feedback, learning and resource supply have distinct
 84 state spaces and assumptions. Section 7 then examines what the observables
 85 can establish, before Section 8 develops the cortical hypothesis and an awak-
 86 ening experiment. The hypothesis is that extracellular contraction increases
 87 field-mediated coupling; its direction, rate and relation to recovery require
 88 independent calibration. Supplemental Material organizes the constructions
 89 and controls behind the claims they support.

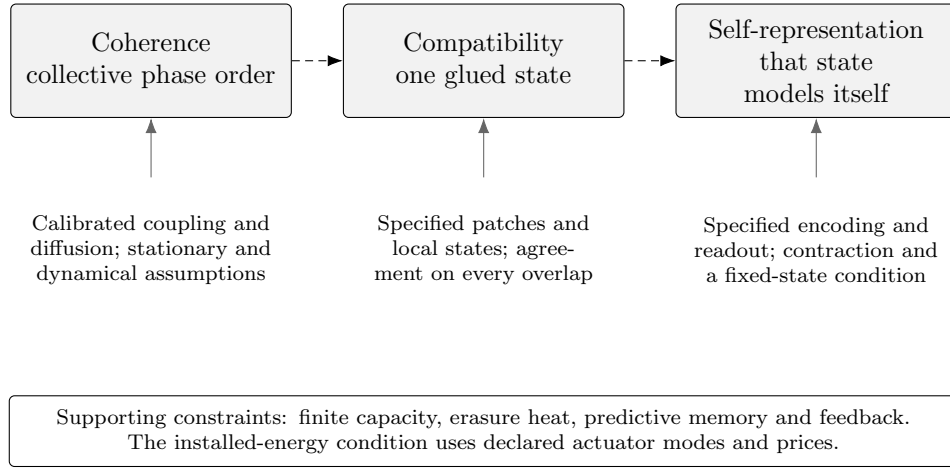


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

90 2 What holds without the cortical hypothesis

91 Most of what follows is conditional. Table 1 names eight hypotheses that
92 the composition consumes rather than derives, and the cortical identification
93 of Section 8 is a further commitment on top of them. Five results are not
94 conditional in that way. They constrain any account that reads unity off
95 phase coherence, whatever physical field is proposed to carry the coupling
96 and whatever substrate carries the field. Each is stated below where it is
97 used; collected here, they are the article’s fixed points.

98 **Phase order alone does not constrain content.** Assign every patch
99 the same phase and let two local mass profiles disagree at a shared site.
100 The profiles are chosen independently of phase, so synchrony bounds their
101 disagreement not at all, and they have no common extension: perfect locking
102 is consistent with local descriptions that do not glue (Section 4). What the
103 positive result of Section 4.2 costs is exactly this gap. A bound on content
104 disagreement needs encoders already related on overlaps, and that relation
105 is a supplied fact about the descriptions rather than a consequence of the
106 dynamics.

107 **The order parameter does not determine the state.** A field winding
108 once across a periodic sheet has $r = 0$ while every patch of it stays locked,
109 and this is not a contrived initial condition. On a periodic array of any
110 dimension, a winding is stationary for *any* isotropic symmetric kernel at

111 identical frequencies, because the coupling term is odd in the separation
 112 against a kernel even in it (`char_is_kuramoto_trajectory`). Section 7.3
 113 exhibits a sheet on which uniform and winding initial conditions reach 0.9657
 114 and 0.0083 at one coupling, diffusion and kernel. A low resultant therefore
 115 reports either weak coupling or a winding field, and no measurement in r
 116 alone separates them. The state also carries an integer the resultant does
 117 not: read around a ring, the same winding has winding number q , and a
 118 nonzero q obstructs every global real-valued phase lift (`winding_degree_`
 119 `obstructs`). The resultant is an average over sites and moves when they are
 120 reweighted or the patches redrawn; the degree telescopes and does not.

121 **A unique fixed point is not a self-model.** A constant readout con-
 122 tracts, fixes whatever state it returns, and reconstructs nothing. Uniqueness
 123 is consequently the wrong criterion, and Section 5.2 gives the right shape
 124 of one: reconstructing M states pairwise more than 2ε apart within error ε
 125 requires M distinguishable codes, so an alphabet of 2^b codes caps the family
 126 at $M \leq 2^b$. This is a counting argument and holds for any encoder and
 127 readout whatever.

128 **Where the encoding is a restriction, the region must resolve the**
 129 **family.** Read the code as the state's own restriction to a region and
 130 the criterion moves off the readout entirely. A boundary whose region
 131 reads two relevant states identically cannot reconstruct them more than 2ε
 132 apart (`not_isRestrictionResonance_of_reconstructs`), and a resonant

133 boundary that does reconstruct a separated family restricts injectively on
134 it (`restrictToAvatar_injOn_of_isRestrictionResonance`). Where the
135 region sits is fixed before an encoding is chosen, so no readout supplies
136 resolution the region lacks.

137 **Agreement after a change takes communication.** In a synchronous
138 network whose updates read only the site’s own state and its declared neigh-
139 bours’ messages, two configurations agreeing on a site’s causal past through
140 T rounds leave that site identical at round T . Two interventions outside
141 that past that require reports more than 2ε apart therefore cannot both be
142 reported from it within ε by the deadline (Section 5). Snapshot compatibil-
143 ity and timely compatibility are therefore different requirements, and cutting
144 the path removes the second while leaving the first intact.

145 None of these depends on the coupling being electromagnetic, on the sub-
146 strate being cortex, or on the eight hypotheses of Table 1 holding. Section 9.3
147 applies four of them — all but the one about the resultant, which is a state-
148 ment about a spatial observable — to a candidate that is neither biological
149 nor field-coupled.

150 **3 Coherence: physical coupling and its re-** 151 **source condition**

152 **3.1 Dynamics and the stationary threshold**

153 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

154 Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and
155 K_{ij} has inverse-time units, with angles dimensionless. Uniform mean-field
156 coupling is $K_{ij} = K/N$; frequency heterogeneity and diffusion are distinct
157 parameters. The stationary calculation below uses identical frequencies and
158 this scalar coupling. Applying it to a spatial kernel requires an aggregation
159 rule and a check of the resulting approximation.

160 For identical frequencies in their rotating frame, the positive normalized
161 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

162 where I_n are modified Bessel functions and ψ is the collective phase. The for-
163 mal derivation uses the absence of stationary current for a periodic gradient
164 drift to obtain this density, then verifies normalization and self-consistency.
165 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988). Above

threshold, exactly one positive solution coexists with zero and increases
strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

Figure 2 shows this stationary branch.

Linearization about the uniform density gives rate $K/2 - D$ for the first harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists exactly above threshold. About the coherent density, the quadratic form is strictly dissipative transverse to collective rotation, whose rate is zero. These linear estimates concern the limiting equation. Relating them to finite stochastic trajectories requires a dynamical mean-field limit and branch selection; exchange symmetry alone does not establish the needed independence (Sznitman, 1991).

3.2 Installed energy bounds mean coupling

A spatial model replaces the finite sum by integration against a kernel. One declared implementation carries response modes with continuous profiles ϕ_i , installation prices u_i and nonnegative occupancies $c_i(z)$ in configuration z :

$$K_z(x, y) = \sum_i c_i(z) \phi_i(x) \phi_i(y), \quad U(z) = \sum_i u_i c_i(z). \quad (4)$$

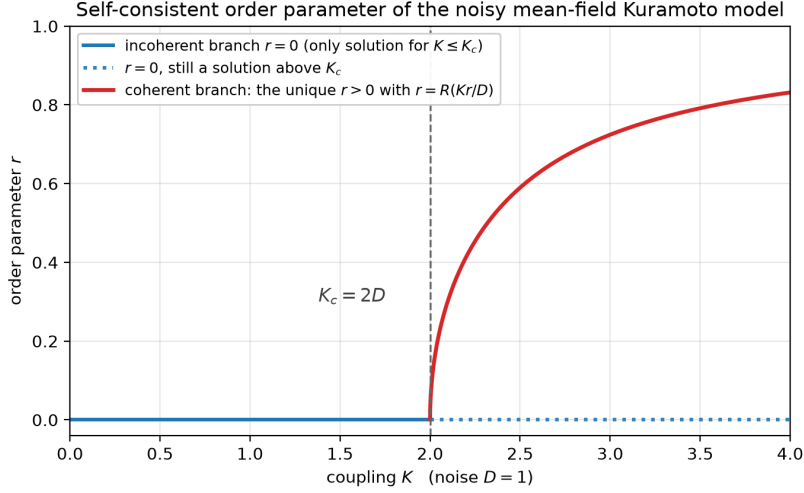


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

181 The same configuration sets the reciprocal kernel and stored energy. An
 182 executed transition changes the kernel through its changed occupancies and
 183 requires work $U(z') - U(z) + Q$, with Q positive into the reservoir. The
 184 installed kernel is averaged over the resulting configuration law $p(z)$. Cell-
 185 pair reconstruction on refining spatial partitions converges to its product-
 186 measure integral under the specified compactness and sampling assumptions;
 187 the refinement index is independent of the number of control updates.

188 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 189 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's

190 integral gives

$$\begin{aligned} U_{\text{inst}} &= \sum_z p(z)U(z), \\ K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_z p(z) \sum_i c_i(z) m_i^2 \leq \kappa U_{\text{inst}}. \end{aligned} \tag{5}$$

191 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
192 coherent stationary branch and any growing Fourier mode about the uniform
193 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
194 is necessary: energy above it need not produce sufficient coupling because
195 the bound can be loose.

196 The quantity constrained is installed energy rather than dissipated heat,
197 and how much the condition excludes depends entirely on κ , which the mode
198 profiles, prices and substrate normalization supply: any macroscopic sub-
199 strate carries installed energy far above the thermal floor, so the inequality
200 binds through that conversion and not through the energy, and no result
201 here derives κ for cortex or for any other candidate. What the condition
202 contributes is a resource requirement in units a measurement can reach. Ap-
203 plying the scalar threshold to heterogeneous or signed spatial dynamics re-
204 quires a justified reduction, and identifying the installed kernel with a cortical
205 field is a separate physical commitment.

206 4 Compatibility: from coordinated activity 207 to a global description

208 4.1 Why common timing is insufficient

209 For overlapping patches U and V , compatibility means that restricting either
210 local description to $U \cap V$ gives the same object. Descriptions can agree on
211 shared quantities while contributing different information elsewhere. Com-
212 patible sections of the specified sheaf have a unique global extension. On
213 a finite cover, local finite spatial measures agreeing as measures on every
214 overlap also have a unique finite-measure extension whose sheaf image is the
215 glued section. No mass-one normalization is imposed.

216 In the cup example, visual and reaching descriptions must express a
217 shared quantity, such as cup location, in common coordinates. Restriction
218 retains that quantity while dropping details particular to each description.
219 Agreement permits surface appearance and hand configuration to contribute
220 to one situation. The mathematical counterexample assigns the same phase
221 to every patch while two local mass profiles disagree on a shared site. Those
222 profiles have no common extension. Because they are chosen independently
223 of phase, synchrony places no constraint on their disagreement.

224 This distinction limits the biological interpretation. Receptive territories
225 and content domains must be related to spatial patches, and the descriptions
226 must support the required extension. Matching overlap marginals of de-

227 coded probability laws need not determine a unique global joint distribution
 228 (Abramsky & Brandenburger, 2011).

229 The cover is itself a choice, and the construction constrains it in one way
 230 only: synchronized patches agree where they overlap. Where they do not
 231 overlap there is nothing to compare, so a cover by pairwise disjoint patches
 232 satisfies that obligation for *every* assignment of local descriptions and phases
 233 (`ofDisjointCover`) and the gluing theorem still returns a unique global sec-
 234 tion, which is the tuple it was handed (`glue_unique_of_disjoint`). Member-
 235 ship of the class then reports nothing about the system beyond the patches
 236 covering it. What a gluing claim asks of a cover is therefore a non-degeneracy
 237 the theorem does not assume: that distinct patches meet (`HasNonempty-`
 238 `Overlaps`), and that what they share is a quantity fixed independently of
 239 the agreement it is being used to establish.

240 4.2 When coherence does constrain content

241 A declared relation between phase and content gives a positive result. Let
 242 contents take values in a real normed vector space E . Patch i reads an
 243 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 244 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 245 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 246 sites; their equality on overlaps is a supplied relation between descriptions of
 247 the shared quantity. The order-parameter identity $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$
 248 implies $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$ for each pair. Consequently every

249 overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2} N\sqrt{1-r^2}. \quad (6)$$

250 Here discrepancy is the norm of the difference, and r and N are read on
 251 whichever population the bound is applied to. That choice is what decides
 252 whether the estimate says anything. A patch carries an order parameter of
 253 its own, and reading the bound there carries that patch's resultant and that
 254 patch's site count (`chord_le_of_patch_coherence`, `compatible_of_patch_`
 255 `coherence`), so a large population observed through small patches is con-
 256 strained by the small count. Read instead on the whole array, the factor N
 257 makes the estimate weak, because it draws a pointwise guarantee from an
 258 average over N^2 pairs. At $r = 1$ the observations agree exactly at either
 259 scale. A shared scalar encoder is a special case. Dropping the overlap rela-
 260 tion permits disagreement even at perfect locking. A common nonexpansive
 261 linear projection preserves agreement, but agreement of a projection does
 262 not establish agreement of the full vector.

263 What Eq. (6) bounds is a chord, which never exceeds 2, so the estimate
 264 constrains anything only where $\sqrt{2} N\sqrt{1-r^2}$ falls below that. The winding
 265 sweep of Section 7.3 fixes both ends of that range on one sheet. Read on the
 266 sweep's own patch, 625 sites at patch-local order 0.9015, the bound is $382.5 L$
 267 and constrains nothing; holding that order fixed, the count alone brings it
 268 below the maximum chord only at 3 sites or fewer. A nearest-neighbour pair

269 of the same winding, the smallest patch there is, gives $0.1388 L$. Shrinking
 270 patches therefore thins the overlaps at the same rate as it tightens the bound,
 271 and the two ends are not reached by the same cover. At a columnar grain
 272 the estimate constrains agreement inside one locally synchronous population
 273 and is vacuous across the distant territories whose agreement unity is about.

274 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 275 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 276 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 277 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

278 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 279 duce an initial mismatch. Approximate observations leave a nondecaying
 280 term in the error bound. Taking a uniform bound on Eq. (6) connects this
 281 update to the phase process.

282 4.3 What approximate agreement determines

283 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
 284 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
 285 Weights that vanish outside their patches and sum to one at each site select
 286 a global profile by weighted averaging. The selection lies within ε of each
 287 patch there, and any two such selections differ by at most ε , independently
 288 of cover multiplicity. An L -Lipschitz readout consequently varies by at most

289 $L\varepsilon$ across selections.

290 At zero tolerance, the selection is the unique exact extension. At every
291 positive tolerance there are families with no exact extension; selection then
292 gives an approximation whose dependence on weights is bounded. Supple-
293 mental Material gives a three-site witness attaining the bound. The mass-
294 profile construction uses the scalar case of the content dynamics and addi-
295 tionally requires nonnegativity of the initial profiles and observations. The
296 general vector result supplies no identification with the spatial-measure sheaf.

297 These constructions concern declared objects: a specified cover, contents
298 valued in a normed space, and encoders related on overlaps. They select
299 neither the neural cover nor the decoder, and identifying their contents with
300 perceptual content, or establishing that the identification bears on experi-
301 ence, is a further modelling task; Section 7 sets out what a measurement of
302 it would have to establish.

303 **5 Self-representation: a state that models it-** 304 **self**

305 **5.1 The reflexive construction**

306 Compatibility gives a global state. Reflexivity relates it to an encoding within
307 that state. Equip global sections with a complete metric. If the additional
308 restriction-resonance condition identifies encoding with restriction to a region

309 U , the reconstruction map is

$$F(s) = \text{readout}(s|_U). \quad (8)$$

310 The region and readout are fixed across the state space. If F contracts and
 311 fixes the section s_* formed from the local descriptions, Banach's theorem
 312 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89
 313 supplies the contraction and $F(s_*) = s_*$; restriction resonance is separate.
 314 Supercritical coupling makes the stipulated factor $q = e^{-(K-2D)\tau/2}$ less than
 315 one for $\tau > 0$. Constructing a map with that factor remains a modelling
 316 obligation.

317 In the cup example, the global state includes the represented scene and the
 318 perceiver's relation to it. A local encoding must retain enough information
 319 for the readout to recover that modelled state; cup position alone would be
 320 insufficient if hand configuration and viewpoint could not be reconstructed.

321 For the same map with Lipschitz factor $q < 1$, approximate reconstruction
 322 gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - q}. \quad (9)$$

323 At the stipulated rate this radius grows as $2\varepsilon/((K-2D)\tau)$ near threshold, for
 324 fixed positive ε and τ . Exact uniqueness survives while the error guarantee
 325 weakens.

326 5.2 Reconstruction across a family of states

327 Sensitivity can be required quantitatively. Declare relevant macrostates, an
 328 encoder E and readout R , and require $d(s, R(E(s))) \leq \varepsilon$ throughout that
 329 family. If two states have the same code, the triangle inequality gives

$$E(s) = E(t) \implies \max\{d(s, R(E(s))), d(t, R(E(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

330 Thus M relevant states pairwise more than 2ε apart require at least M
 331 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
 332 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
 333 existing self-prediction map.

334 This is an additional accuracy criterion on a family, with no requirement
 335 for multiple exact fixed points. A blind encoder or constant readout fails it
 336 for separated states despite having a unique fixed point.

337 Where the encoding is read as the state's own restriction to a region, the
 338 criterion becomes a condition on the region. Restriction resonance requires
 339 the encoder and that restriction to identify the same pairs of states, so a
 340 boundary reconstructing two relevant states more than 2ε apart while its re-
 341 gion reads them identically is not resonant (`not_isRestrictionResonance_`
 342 `of_reconstructs`); read forwards, a resonant boundary that reconstructs
 343 a pairwise separated family restricts injectively on it (`restrictToAvatar_`
 344 `injOn_of_isRestrictionResonance`). Which states a region reads identi-
 345 cally is fixed by where the region sits, before an encoding is chosen, so the

346 resolution the family demands is the region's and no readout supplies it. The
 347 implication runs one way: a region that does resolve the family is not thereby
 348 shown to carry a resonant encoding, and E89 supplies no such resolution.

349 Those two demands are jointly satisfiable with nothing folded anywhere.
 350 Take the region to be the whole substrate: restriction is then the identity, so
 351 the boundary is resonant, reconstructs every declarable family exactly, and
 352 restricts injectively on all of them (`topAvatar_satisfies_obligations`). A
 353 properness condition on the region (`IsProperAvatar`) is therefore a separate
 354 requirement rather than a consequence of resonance and resolution, and what
 355 a fold into a local region is asked for. Its quantitative form replaces the
 356 alphabet by a metric. With an L -Lipschitz readout, a family reconstructed
 357 within ε whose members lie pairwise more than r apart has δ -separated codes
 358 whenever $\delta L \leq r - 2\varepsilon$, so the family is no larger than the δ -packing number
 359 of what the region carries (`encard_le_packingNumber_restrictToAvatar`).
 360 That is the shape a continuous family of states needs, and at the whole
 361 substrate it says nothing, because the region's range is everything. The gain
 362 L is declared and derived here for no physical readout, and at unbounded
 363 gain the bound says nothing.

364 5.3 Agreement after a change takes communication

365 Snapshot compatibility does not specify how quickly descriptions respond
 366 when a shared quantity changes. In a synchronous network whose updates
 367 read only the site's own state and declared neighbours' messages, two initial

368 configurations agreeing on a site’s causal past through T rounds produce the
 369 same state there at round T . The deadline consequence is the one Section 2
 370 states, and it follows from the reconstruction bound in Section 5.2: a report
 371 from that site is a readout of a code that the two interventions leave identical.

372 A three-site delay line witnesses failure after one round and exact report-
 373 ing after two; cutting the path prevents delivery at every round. A report
 374 may still happen to be correct about one fixed world. The bound concerns
 375 responsiveness across interventions.

376 Throughout this section *prediction* names a reconstruction map, and the
 377 relevant family of states, the metric, the tolerance, the communication graph
 378 and the deadline are all declared inputs, as are the represented variables and
 379 the fidelity counted as accurate. Reading the map as temporal neural pre-
 380 diction would require dynamics; reading the resulting code count as physical
 381 memory capacity would require a hardware model, which a state-dependent
 382 readout would have to extend.

383 6 Thermodynamics and composition

384 The phase and representation results concern coupling, content disagreement
 385 and reconstruction error. Thermodynamics constrains the physical opera-
 386 tions that could implement them, and it does so from the side rather than
 387 along the spine: no edge below derives coherence, compatibility or reconstruc-
 388 tion from a heat bound, and the resource condition these edges terminate in

389 is necessary without being restrictive for any macroscopic candidate (Sec-
 390 tion 3.2). What this half contributes is a ledger in which a resource claim
 391 is stated in checkable units. Finite-register capacity obeys $H(p) \leq \log |X|$;
 392 erasure heat depends on specified system–bath assumptions (Landauer, 1961;
 393 Norton, 2005; Shenker, 2000). A separate boundary result concerns a field
 394 reaching disconnected vacuum components (Kibble, 1976). The composition
 395 connects these objects through the explicit hypotheses in Table 1.

396 **6.1 Feedback changes the information constraint**

397 For passive memory X_t driven by an environmental signal S_t , the predictive
 398 bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (11)$$

399 The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still
 400 et al., 2012). Zero retained information also satisfies it, so a task and a
 401 learning rule require additional specification. Relating this bound to Lan-
 402 dauer heat requires a common thermal scale and budget, and memory waste
 403 bounded by erased entropy.

404 An acting agent changes the signal it will receive. Specify policy, world,
 405 observation and memory-update channels. For their finite laws,

$$\begin{aligned}
 I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\
 &\quad - I(X_t; S_{t+1} \mid S_t).
 \end{aligned} \quad (12)$$

406 The second conditional term vanishes in the passive model. With feedback,
 407 the signed difference can be negative as action creates correlations with the
 408 future. A causal reading requires an environmental state sufficient to describe
 409 the relevant dynamics.

410 For a controlled substep, the active thermodynamic bound instead uses
 411 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F \| P_R) =$
 412 $\Delta H(X, S) + \langle Q \rangle / \theta \geq 0$ under local detailed balance, with $\theta = k_B T > 0$ and
 413 heat positive into the reservoir (Horowitz and Esposito, 2014; Ito and Sagawa,
 414 2013). Every forward path of positive probability must have a positive-
 415 probability reverse for this finite identity. A missing reverse gives infinite
 416 divergence. The supplement develops the support conditions and composes
 417 actuation with memory updates on their shared intermediate law.

418 **6.2 Learning and repeated operation need explicit re-** 419 **sources**

420 The finite examples distinguish selecting a feasible policy, relaxing a register
 421 toward an encoded reward, and learning an unknown task from observa-
 422 tions. The observational learner acquires correlation with an environmental
 423 parameter while its action marginal stays uniform. A continuing model exe-
 424 cutes actuation, observation-driven learning and reset in sequence, preserving
 425 above-chance performance at a positive work cost per cycle.

426 A phase-derived sensor makes the informational distinction concrete: an

427 informative readout and a blind control have the same phase order but dif-
 428 ferent learning performance. Attaching the learner’s register to the actuator
 429 of Section 3.2 ties reward, installed kernel and work to its own executed law.
 430 The learning witness rises above the necessary installed-energy bound while
 431 its matched control remains on the exclusion line. These finite constructions,
 432 detailed in Supplemental Material, leave the evolving feedback between phase
 433 dynamics, installation and sensing to be specified.

434 For that model, common energy E , thermal scale θ , local detailed balance
 435 and positive support at each stage give

$$\theta[H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle E \rangle_0 - \langle E \rangle_N \leq B + \langle E \rangle_0 - \langle E \rangle_N, \quad (13)$$

436 where expected cumulative work stays within the supplied allowance B at
 437 every prefix. This bounds expectations. Separate constructions track a store
 438 on each trajectory, include finite sources, charge sensor clearing and prepend
 439 preparation. They specify when a mean allowance represents an available
 440 resource; an expected budget alone permits overdrawn paths.

441 These constructions establish costs for their declared processes. A finite
 442 source with a positive net cost permits only finitely many funded cycles,
 443 while continuing operation requires replenishment. Register–bath witnesses
 444 also distinguish actual bath-energy transfer from transition-log heat. Iden-
 445 tifying a work store, an erasure budget and the installation energy of Sec-
 446 tion 3.2 requires physical relations that these separate models do not supply

447 automatically.

448 **6.3 What the composition establishes**

449 The passive theorem `chain` and feedback theorem `chain_active` each ac-
450 cept eight connecting implications (Table 1). In the active branch, a named
451 substep at the register’s thermal scale satisfies

$$\theta > 0, \quad \theta[H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (14)$$

452 with B the register’s heat. The upper allocation follows from a ledger whose
453 operations deliver that total heat to one reservoir and whose other operations
454 do not increase joint entropy. Those ledger assumptions are physical inputs.
455 Neither Landauer’s lower bound nor microscopic reversibility supplies them.
456 The regime in which those premises are exhibited is a near-equilibrium one:
457 strictly positive forward and reverse elementary paths, local detailed balance
458 at one thermal scale, and entropy differences and heats of order $k_B T$, the
459 witnesses being two-bit systems with interaction energies of that size. Cortex
460 is not in it. The installed energy of Section 3.2 is 2.4×10^{14} times kT at body
461 temperature over one region and one phase-diffusion time, and a budget B
462 of that size stands against an entropy term of order $k_B T$, so Eq. (14) is
463 satisfied by any macroscopic candidate and excludes none of them. Reading
464 the register instead at the scale where its premises are physical, such as a
465 single channel, relocates the difficulty to E45Active, which would then have to

466 carry one molecular step’s budget to the convergence of a continuum kernel.
467 The active branch is an alternative route to the same node rather than a
468 stronger one, and no result of Section 3.2 passes through it: the installed-
469 energy condition is derived from E56 and E67.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12 <i>capacity and boundary</i>	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.
E23 <i>boundary to register</i>	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.
E34 / E34Active <i>register to dissipation</i>	Bridge assumption	Passive: predictive memory at the register’s thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations’ compressiveness. Both are exhibited at thermal-scale heats on finite state spaces with positive reverse paths, which is not the regime a cortical register occupies.
E45 / E45Active <i>couplings to continuum</i>	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Convergence follows from mesh refinement rather than from the budget, which selects no regime at macroscopic scale. Refinement and physical identifications remain inputs.

Table 1 (continued)

Connection	Status	Required assumption
E56 <i>field identification</i>	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67 <i>above threshold</i>	Physical commitment	That field operates above the synchronization threshold.
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements. Patches and oscillators carry one index, so one oscillator carries exactly one content patch: a modelling assumption rather than a consequence, and not an innocent one, since field-mediated coupling is spatial while two territories describing one quantity need not be neighbours.
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $q < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 3.2’s occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

Both branches also take a thermodynamic cover carrying exact overlap compatibility and a reflexive boundary specifying encoding and readout. They conclude **UnifiedSelf**: the glued section is the unique fixed point of self-prediction. Compatibility and accuracy at that section enter as premises; uniqueness follows. Identifying encoding with restriction needs the separate

475 resonance condition of Eq. (8).

476 Each of the eight edges is supplied rather than derived from the one above
477 it. E12 is equivalent to its conclusion, so capacity and boundary are inde-
478 pendent roots rather than a step from one to the other; both E45 discharges
479 take their convergence from refinement data and not from the thermody-
480 namic bound; E78 states that coherence relaxes the supplied cover, and the
481 coherence-to-content estimate of Eq. (6) bounds disagreement only for en-
482 coders already shared on overlaps. The inferential content of the composition
483 is its last step, where contraction and a fixed glued section give uniqueness.
484 What composing them establishes is that the premises are consistent, named
485 in one place and machine-checked together, and not that coherence follows
486 from installed energy or self-representation from coherence.

487 Joint witnesses combine a register, boundary, refining mesh and small
488 spatial cover; active witnesses add controlled thermal updates and a priced
489 installation. They establish satisfiability under their stated identifications.
490 Mesh convergence follows from refinement data, and the witnesses establish
491 no common cortical mechanism. Supplemental Material maps the claims to
492 their Lean declarations and records the counterexamples. The development
493 declares no additional axioms beyond its libraries' foundational axioms.

494 7 What the observations establish

495 The composition holds under its premises. Whether observation can estab-
496 lish those premises is a separate question, and the three requirements it
497 connects concern different objects: phase dynamics, local descriptions and
498 internal encodings. Their observables must retain these distinctions. The
499 following controls show which mechanisms remain indistinguishable, which
500 temporal approximations fail, which states one observable cannot separate,
501 and what independent content measurements must establish. Each is a finite
502 numerical regime on a declared model, with full parameters and controls in
503 Supplemental Material, and none of them calibrates a cortical quantity.

504 7.1 Stationary distributions leave the mechanism un- 505 determined

506 Equation (2) gives a concentration–coherence relation with no fitted shape
507 parameter. Estimate r from circular resultant length and a from density
508 shape: inverting R from the measured r would make agreement automatic.
509 Even independently estimated agreement identifies a statistical family shared
510 by different mechanisms.

511 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
512 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
513 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
514 oscillators with prescribed drift $-h(t)\sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,

515 and diffusion D_0 have that same density. Their stationary distributions co-
 516 incide window by window despite different causes.

517 Resolved dynamics supply additional information. In the ideal model with
 518 continuous paths, quadratic variation accumulates at rate $2D$ and conditional
 519 increment means constrain drift. Separating endogenous coupling from a
 520 matched common drive additionally requires a measured drive or controlled
 521 perturbation; identifying electromagnetic coupling requires evidence specific
 522 to that interaction. Supplemental Material gives the equivalence construction
 523 and its numerical check.

524 **7.2 Finite ramps change the observable onset**

525 The square-root law in Eq. (3) describes stationary solutions. A changing
 526 coupling requires time for the phase distribution to approach those solutions.
 527 In a deterministic mean-field control that crosses the threshold symmetrically,
 528 none of the scanned rates meets the declared tracking tolerance: slowing
 529 the crossing does not remove the terminal residual under that initialization.
 530 A slow control variable therefore supplies no general guarantee of branch
 531 tracking at onset. Supplemental Material gives the residual, rate range and
 532 admissible rates away from threshold.

533 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas
 534 at 8 coupling speeds from $v = 0.1$ to $v = 0.0001$. Escape is the first of three
 535 consecutive saved samples at $r \geq 0.2$ after threshold. Faster ramps delay
 536 escape (Fig. 3); only 5/32 replicas escape within the fastest window. The

537 6 uncensored legs supply the delay scaling below; the decade-spaced ones
538 among them have mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted
539 onset exponents differ from those of the exact stationary branch evaluated
540 on the same grids. At the fastest speed the ensemble mean never reaches the
541 fit window, so its fitted slope does not estimate stationary onset.

542 Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002)
543 makes the apparent exponent depend on ramp speed and fitting window.
544 The specified deviation-based critical ramp speed is not bracketed by this
545 sweep. With fluctuating coupling, the relevant stationary comparison also
546 depends on rate: the mean coupling approximates order in the tested fast
547 regime, while averaging stationary order over coupling works for the tested
548 slow drives confined to one side of threshold. Slow crossing drives can satisfy
549 neither approximation.

550 **The prediction.** The delay scaling is what this section supports as a pre-
551 diction, and it needs no absolute calibration. Over the 6 uncensored legs the
552 mean delay follows $\Delta K \propto v^{0.444}$, with a 95% slope interval of [0.394, 0.494],
553 which does not contain 1/2. What that shortfall measures is the escape cri-
554 terion and not the scaling. Applied to a deterministic mean-field run seeded
555 at the fluctuation floor, the same estimator returns 0.425 over the same legs
556 at the same escape criterion, inside the interval on the measurement. That
557 run has no noise to delay it, so the shortfall is the criterion's: it crosses the
558 saturation between the floor it starts from and the level it is stopped at, and

559 tightening the criterion to $r \geq 0.05$ returns 0.490. The bias is reproduced
560 where the answer is known, which is what licenses reading the measurement
561 as the law rather than as a competing exponent.

562 Stated as a law rather than as a number: onset delay scales as the square
563 root of the rate at which coupling crosses threshold. Testing it requires only
564 that $\dot{K}$ be monotonic in a manipulated rate — not its value, nor κ , nor γ
565 — so it is a within-subject design that varies the rate of emergence from
566 anaesthesia alone, with each participant as their own control. Such a design
567 must avoid the hazard this sweep exhibits: a fixed absolute escape criterion
568 meets a size-dependent critical fluctuation floor, so the criterion has to be
569 referred to each recording’s own pre-threshold fluctuation scale rather than
570 to a shared number. The diffusion D is the remaining input, and Section 7.1
571 notes that it is estimable from resolved phase traces through the rate $2D$
572 at which quadratic variation accumulates, which is a different quantity from
573 the concentration the propofol exercise could not range over.

574 Additional controls address integration bias, population size and spatial
575 or signed coupling; the refined signed-network crossing bracket $K_{\text{eff}}/D \in$
576 $[1.958, 2.009]$ contains 2.

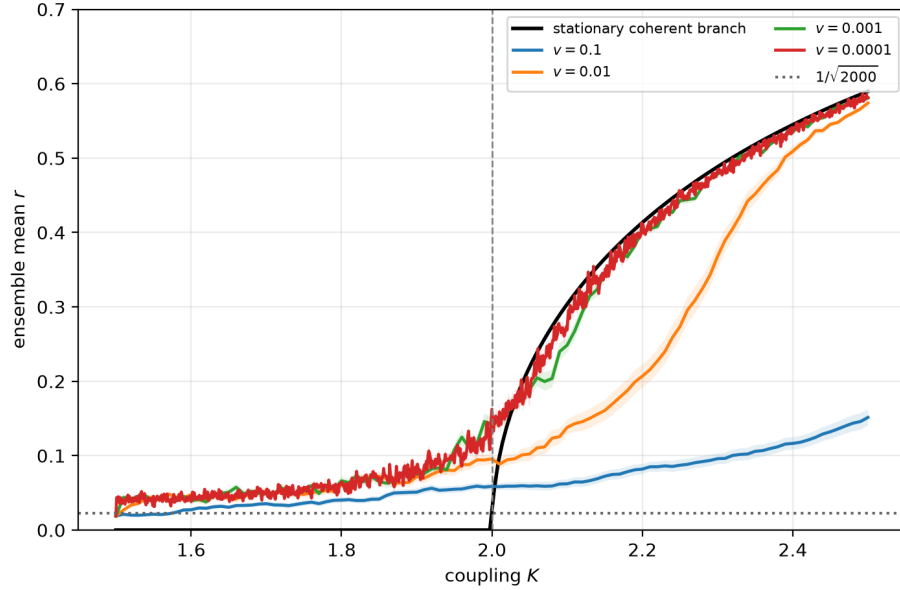


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

577 7.3 A winding state leaves the order parameter am- 578 biguous

579 The order parameter r is indifferent to where a phase sits in space. A field
580 winding once across a periodic sheet therefore has $r = 0$ while every patch
581 of it stays locked. Such a field is not an artificial construction here. On a
582 torus a winding is stationary for any isotropic symmetric kernel at identical
583 frequencies: the coupling term is odd in the separation against a kernel even
584 in it, so the drift cancels pair by pair. `char_is_kuramoto_trajectory` proves

585 this for any finite periodic array, this sheet included. A winding satisfies the
586 positivity and identical-frequency hypotheses. It violates only the quarter-
587 turn initial spread that the convergence result additionally requires before
588 placing the limit at the uniform state.

589 On the sheet of Section 3's spatial control, a winding survives below
590 an interpolated decay length of 0.1871 mm and unwinds above it (Fig. 4).
591 That boundary is interior to the 0.1–0.3 mm band. At 0.1 mm the same
592 sheet at the same coupling, diffusion and kernel reaches $r = 0.9657$ from a
593 uniform start and $r = 0.0083$ from a winding one. Both are stationary, with
594 patch-local order 0.9015 and no winding defect. At fixed K the observable
595 therefore takes two values spanning nearly its whole range. Coupling does
596 not determine the resultant, and a low resultant does not by itself distinguish
597 weak coupling from a winding field.

598 Under the quenched frequency spread of 1.5 rad s^{-1} that the spatial con-
599 trol carries, the boundary moves to 0.1421 mm. The same comparison at
600 0.1 mm gives 0.9418 against 0.0348, holding across 3 frozen realizations.
601 Heterogeneity shortens the range that sustains a wave without removing it.

602 Which windings persist has a criterion behind it. Linearizing the phase
603 model about a winding returns a coupling of the same translation-invariant
604 form, each separation reweighted by the cosine of the phase the winding
605 advances across it (`hasDerivAt_charJacobian`). That reweighted kernel is
606 again even in the separation, so the same characters diagonalize it and ev-
607 ery mode's rate is one of its eigenvalues. Where the winding advances by

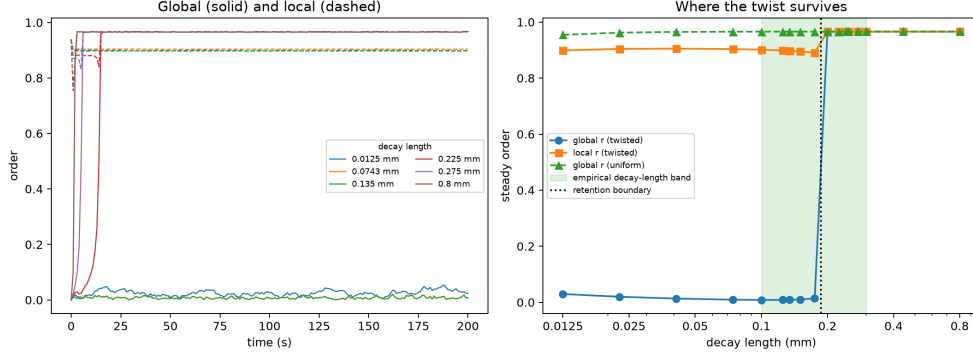


Figure 4: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

608 at most a quarter turn across every separation the coupling reaches, the
 609 reweighted kernel is nonnegative and no rate is positive (`charJacobian_`
 610 `stable_of_quarter_turn`); where some rate is positive, that mode grows
 611 without bound (`charJacobian_growing_mode`). Moving coupling mass to
 612 longer separations is what carries a phase advance past a quarter turn, and
 613 that is the mechanism the decay-length sweep varies. The criterion bounds
 614 rates rather than trajectories: asymptotic stability is not proved, so the
 615 states that persist occupy a narrower range than it admits, and the bound-
 616 ary quoted above is measured and not derived from it. The same kernel is the
 617 potential's curvature, so a winding at $r = 0$ is not the global minimum the

618 convergence result places its limit at, and one carrying a growing mode is not
619 a local minimum either: that result's quarter-turn hypothesis must exclude
620 a winding rather than merely decline to cover it (`char_not_potential_min`).

621 An amplitude layer decides which of these states exist, and leaves what
622 the resultant reports unchanged. Every oscillator of the phase model sits on
623 the unit circle, so a field with an amplitude node has no representation in
624 those dynamics. The real Stuart–Landau extension supplies one on the same
625 coupling, and leaves intact the gradient structure the convergence results rest
626 on. The state becomes $z_i \in \mathbb{C}$ with $\dot{z}_i = (\mu - |z_i|^2)z_i + \sum_j A_{ij}(z_j - z_i)$. Its
627 phase equation is the Kuramoto equation with each coupling weighted by a
628 ratio of amplitudes (`amplitude_phase_is_kuramoto`). The phase model is
629 this one with the amplitude ratios set to one, rather than a separate object.

630 What the added degree of freedom decides is which windings exist at all.
631 On a periodic array of any dimension, this sheet included, a winding requires
632 $\mu + \lambda > 0$. Here $\lambda = \sum_d f_d(\cos \phi_d - 1)$ is the eigenvalue the kernel supplies,
633 summed over separations, with ϕ_d the phase the winding advances across d
634 (`exists_charState_iff`; on the ring $\phi_d = 2\pi qd/n$). For the nonnegative
635 kernels used here $\lambda \leq 0$, and moving coupling mass to longer separations
636 lowers it further. A broad enough kernel therefore removes the winding out-
637 right. The sweep above reaches that same boundary by a different route:
638 the phase model retains every winding at every coupling, and loses this one
639 under integration instead. The two layers bound the same regime from op-
640 posite sides. A phase recording does not separate a state the amplitude layer

641 excludes from one the dynamics fails to sustain.

642 Three limits bound the layer for a reader of the cortical literature. These
643 states are windings and not spirals, and on the sheet that is a theorem rather
644 than a shortage of dimensions: equal modulus at every site puts the phase
645 circulation at exactly zero on every plaquette, and on a periodic sheet the
646 circulations of *any* field sum to zero, so a lone defect is not a state of this
647 geometry at all. Relaxational dynamics has no propagation speed, which is
648 the quantity a cortical wave measurement leads with. And $|z|$ is a normal-
649 form radius, which a measured envelope is not; this work identifies the two
650 nowhere. Supplemental Material gives the sweep, its uniform control, the
651 amplitude layer, the standing wave it admits inside the band, the potential's
652 curvature at a winding and the defect theorems, together with the remaining
653 limits of both.

654 A vanishing resultant is the absence of global agreement and not of agree-
655 ment. Eq. (6) read on a nearest-neighbour patch of a winding gives a dis-
656 crepancy bound of $2\sqrt{2} L |\sin \phi|$, with ϕ the phase advanced across the patch,
657 which falls to zero as the same winding is spread over more sites (`chord_le_`
658 `of_char_patch`), while the global form of the same bound is vacuous at $r = 0$.
659 Encoders shared on overlaps therefore still bound content disagreement in-
660 side a patch of a state that no global observable calls coherent. The winding
661 number itself stays unconnected to content. It is, however, a property of the
662 state rather than of the cover read on it. On a ring the whole turns dropped
663 by the winding's affine local lifts are zero at every site but the wrap, where

664 the residue resets, and they sum around the loop to q ; a nonzero q obstructs
665 every global real-valued phase lift, so one state at one coupling has resultant
666 exactly zero and degree q (`winding_degree_obstructs`). Reweighting the
667 sites moves the first quantity and redrawing the loop's intermediate transi-
668 tions leaves the second where it was. The scope is the ring's: this is a loop's
669 winding and not a spiral, the transitions are those of the affine lifts rather
670 than of an unwrapped measurement, and the integer is identified with no
671 cortical observable.

672 This qualifies what a phase trace can establish. An onset criterion stated
673 in r alone registers a change of basin as readily as a change of coupling.
674 The protocol of Section 8.3 therefore needs a spatial observable beside the
675 resultant.

676 **7.4 Coherent adaptation can lose environmental align-** 677 **ment**

678 An adaptive coupling rule might supply representational structure, but its
679 objective must reward the relevant agreement. For symmetric coupling, pair-
680 wise sine terms cancel in the population mean drift. Minimizing squared drift
681 at that fixed mean therefore rewards equalizing drift across nodes. In the
682 frequency-cluster control, this favors coupling between unlike-frequency clus-
683 ters, whereas the diagnostic environmental template rewards coupling within
684 clusters.

685 The formalization proves descent of the squared-drift objective for a spec-
686 ified gradient flow at fixed phases. Its functional differs from thermodynamic
687 entropy production: at a coherent stationary density the probability current
688 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
689 rial gives the operator, its gradient and the conditions of the descent result.

690 The finite control co-evolves noisy phases and symmetric, nonnegative
691 couplings at fixed total resource. Its template is a diagnostic covariance of
692 latent signals, never a training input. Across three seeds, second-half order
693 remains between 0.9167 and 0.9222, while the within-cluster to between-
694 cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 5).
695 Random-update, permuted-gradient and frozen-kernel controls are reported
696 in Supplemental Material.

697 The immediate post-update objective reduction disappears when aver-
698 aged over complete update intervals. A prespecified grid of 6 rate–cadence
699 combinations contains no case meeting the 5% sustained-reduction criterion
700 while retaining coherence and losing alignment. The finding is coherent adap-
701 tation away from the specified structure, with no demonstrated sustained cost
702 reduction.

703 Kernel alignment and overlap compatibility are different observables.
704 This control addresses acquisition of its designated template under its par-
705 ticular objective. Learning models can explicitly include data in an objective
706 (Corberi et al., 2026); an account of agreement about the cup must specify
707 the content its learning rule is meant to preserve.

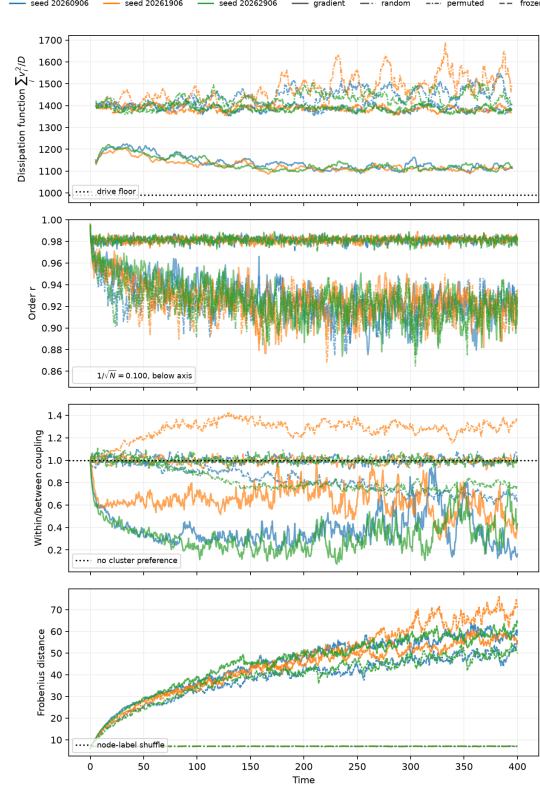


Figure 5: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

7.5 Content agreement needs independent decoders

A neural test needs independently identified domains, local descriptions and restriction maps. Two decoders should estimate a shared content on overlapping domains before their agreement is examined. Discrepancy requires a null under decoding noise and controls using deliberately compatible and incompatible examples. An implementation on constructed data separates these cases at $\text{AUC} = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by trial; the compatible construction scores 0.476 on an interval containing one half. Shared-prior shrinkage can nevertheless manufacture agreement, and regional bias can manufacture disagreement. Per-patch decoding diagnostics detect these failures; the compatibility statistic alone does not.

Phase agreement cannot replace that measurement. On constructed content, phase carries a share 1.000 of the content entropy when content is built from it and 0.000 when it is not. A phase-derived statistic reports compatibility on phase-locked patches whose local descriptions disagree. The implemented content statistic detects their incompatibility. These controls validate a measurement on known constructions; a neural content model and decoder remain to be established.

Choosing the cover also chooses a candidate subject: trivial descriptions can agree while carrying little content, and a partition can conceal disagreement. The useful choice must therefore be constrained by what neural activity encodes. The exact composition assumes compatibility of its supplied cover. The content mechanisms above identify additional hypotheses under

731 which agreement can arise; they leave the empirical choice of content and
732 the connection to that cover explicit.

733 **7.6 What an informative phase-density measurement** 734 **requires**

735 The concentration range and the dependence structure decide whether the
736 concentration-coherence relation can be tested at all. Under the specified
737 log-density estimator and bias-floor criterion, discrimination at the observed
738 concentration ceiling $a = 0.542$ requires 3000 independent sites in the scanned
739 designs, rising to 31000 when sites share a mean in groups of 10. These
740 counts describe that estimator and dependence model; the intended recording
741 requires its own calibration. Supplemental Material reports the bin-count
742 controls and the full dependence-aware design.

743 That ceiling comes from exercising the estimators on public propofol-
744 sedation EEG (Bajwa et al., 2025), pooling channels in 100-ms windows.
745 Bipolar and common-reference montages produce different residuals. The
746 bipolar results are compatible with the Bessel relation over a concentration
747 range too narrow to distinguish it from its linear approximation at the ob-
748 served scatter, which is the range the design scan takes as given. Temporal
749 pooling changes the estimand, and sedation recordings leave the awakening
750 hypothesis untested.

751 An informative dataset therefore needs sufficient concentration range or

precision, calibrated estimators and an observation model for dependence,
source mixing and reference choice. Public mouse calcium imaging provides
thousands of spatially located neurons across waking, sleep and anesthesia
(Oomoto et al., 2026), but its 7.65 Hz frame rate is below the phase-
estimation band used here. Site count, state coverage and bandwidth are
separate requirements.

8 The cortical hypothesis and an awakening test

The cortical proposal identifies a physical source of coupling and asks whether
its changes help explain recovery. The tests in Section 7 determine what
observations must distinguish before a phase trace can support this identification.

8.1 Physical identification and calibration

The cortical hypothesis identifies part of effective coupling with endogenous
extracellular electromagnetic interactions. Fields can modulate spike timing
(Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
propose coordination of distributed neural information (Pinotsis & Miller,
2022; McFadden, 2020). Phase organization under propofol motivates studying
this regime (Bastos et al., 2021), but does not measure coupling.

771 The formal field predicate requires a jointly continuous kernel, a normal-
772 ized substrate, coupling equal to the kernel average, and no single site pro-
773 viding more than half that strength. Toy constructions satisfy it. A cortical
774 realization must specify the scale at which the kernel and its reduction ap-
775 ply: calcium imaging reports different state-dependent modularity patterns
776 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

777 Cortical columns are the obvious candidate unit, and they answer one
778 of the two demands the formalism makes. As the oscillator index they are
779 a serious proposal: they are delineated by response properties and anatomy
780 rather than by the agreement they would be used to establish (Mountcastle,
781 1997), and a phase can be estimated from a column-sized territory at the
782 electrode spacings used for cortical wave measurements (Townsend et al.,
783 2015). As the content cover they are a poor one, for three separate reasons.
784 A parcellation into non-overlapping columns is a disjoint cover, which satisfies
785 the compatibility obligation without constraining anything (Section 4). The
786 cover condition additionally requires the patches to exhaust the substrate,
787 while much of association cortex has no accepted columnar parcellation. And
788 the column’s standing as a canonical cortical unit is contested (Horton &
789 Adams, 2005; Rakic, 2008). Offering columns for both roles at once therefore
790 supplies the oscillator index and leaves the content cover to be identified on
791 separate evidence, which is the bridge assumption E78 records.

792 The spatial reduction is also testable. A row-sum aggregation at zero
793 diagonal tracks the uniform-coupling control over the tested spatial decay

794 range, while frequency heterogeneity displaces the operational threshold. The
795 fit window biases both estimates relative to $2D$; agreement with the control
796 is therefore the supported conclusion. Supplemental Material reports the
797 reduction, its numerical error and the distinct assumptions needed for the
798 kernel-average identification in Eq. (5).

799 The field commitment requires physical evidence beyond a choice of math-
800 ematical representation, and the available support comparisons do not supply
801 it. A finite matrix spread over positive-mass cells preserves its total cou-
802 pling and phase correlation exactly, while point-supported vanishing under
803 an atomless measure and a disadvantage from excluding a better-correlated
804 pair concern specified support constraints; Section 9.3 reads the same com-
805 parisons against a digital candidate. Neither supplies energy prices.

806 A spike-time shift alone supplies no coupling rate. The dimensionless
807 product of population size N , timing sensitivity s , field amplitude E and
808 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N s E f. \quad (15)$$

809 This rate describes how induced timing shifts accumulate into drift. Nei-
810 ther it nor the installation factor κ follows from a heat bound. Testing the
811 physical threshold requires independent estimates of the interaction kernel,
812 its aggregation and diffusion in the same cortical regime.

8.2 Extracellular geometry and recovery

The proposed control variable is extracellular geometry. Extracellular volume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková & Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density, an effective-medium description predicts that a larger extracellular fraction raises conductivity and reduces field amplitude.

The coupling-gated hypothesis consequently predicts that extracellular contraction on awakening raises E and, holding the other factors in Eq. (15) fixed, raises effective coupling. This argument supplies a direction; the geometry–coupling relation, its magnitude and its role in human recovery require measurement.

For a smooth crossing at nonzero speed, evaluating the stationary branch along the changing control gives the benchmark

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (16)$$

This follows from Eq. (3) at fixed D and concerns the stationary branch evaluated at $K(t)$. Reading it as observed recovery requires an independently justified tracking approximation. The crossing controls in Section 7.2 leave that approximation unsupported at onset, even though a separation of geometry and phase time scales can support it away from threshold. The

833 awakening hypothesis therefore needs a calibrated dynamical prediction for
834 an operational onset above the finite-size fluctuation floor. Crossing K_c alone
835 does not supply a recovery latency.

836 **8.3 A protocol for the awakening experiment**

837 The physical test has six requirements:

- 838 **1. Coupling and diffusion in the same units.** Estimate them from re-
839 solved phase dynamics, separating heterogeneity and observation noise.
840 Measure the drive or perturb the system to distinguish endogenous cou-
841 pling.
- 842 **2. Geometry calibrated to coupling.** Validate the geometry observ-
843 able and its relation to effective coupling, with uncertainty. The con-
844 ductivity argument supplies only the expected direction.
- 845 **3. Calibrated phase measurements.** Estimate r by resultant length
846 and a by unweighted regression of log density on centered cosine.
847 Choose site count and binning from a dependence-aware calibration
848 over the intended concentration range. The tested designs require
849 thousands of sites at the observed concentrations; temporal pooling
850 changes the spatial estimand.
- 851 **4. An independent recovery endpoint.** Measure behavioural recovery
852 separately from phase organization.

853 **5. A predeclared dynamical onset criterion.** Set a coherence thresh-
854 old above the finite-size floor and evaluate its latency under the mea-
855 sured coupling trajectory, including ramp speed or fluctuations. Record
856 a spatial observable beside the resultant, since a winding field reports
857 a low resultant at coupling that sustains local order (Section 7.3).

858 **6. Matched model comparison.** Compare coupling increase, diffusion
859 change, common drive and alternative recovery models under the same
860 observation and filtering assumptions.

861 Failure of the concentration–coherence relation over an informative range
862 constrains the density model. Failure of the calibrated geometry–coupling
863 association constrains the physical gate. Better prediction of independent
864 recovery by a competing mechanism counts against its proposed role. A
865 favorable outcome still leaves the content of coordinated activity to be ex-
866 plained.

867 **9 Discussion**

868 Two things have been established here, and they are of different kinds. The
869 constraints of Section 2 hold outright: synchrony does not deliver compatible
870 content, the resultant does not determine the state, a unique fixed point is
871 not a reconstruction, a region that cannot resolve a family cannot encode it,
872 and agreement about a change takes communication. Any account that reads
873 unity off phase coherence has to meet them, this one included. The account

874 set out in Section 1 is conditional at every step by contrast: each connection
875 in it is a hypothesis supplied rather than a result derived, and Table 1 names
876 all eight. Its four non-standard ingredients, named in Section 1, are the
877 installed-energy condition, the patch-local form of the content bound, the
878 criterion separating a winding field from weak coupling, and the machine-
879 checked composition with its joint witness. What follows takes the conditions
880 in turn, as commitments the proposal incurs, as measurements that would
881 bear on them, and as questions the framework leaves open.

882 **9.1 Unity and a minimal self as empirical commit-** 883 **ments**

884 We propose the state glued from compatible local descriptions as a corre-
885 late of conscious unity, and its accurate internal reconstruction as a correlate
886 of a minimal self. The motivations are that different contents belong to
887 one experienced situation and that the situation is experienced from a per-
888 spective. The conditional results identify relations a realization would need:
889 sufficient installed resources for the specified phase model, agreement about
890 shared content, and an encoding that supports reconstruction at the required
891 fidelity. Discharging E12–E89 for cortex would make the composition a state-
892 ment about cortex: that a cortical field above threshold relaxes to a cover
893 whose compatible local descriptions glue, and that the glued state is the
894 unique fixed point of a reconstruction map realized there. It would leave the

895 two identifications above exactly where they are, because neither is among
896 the eight and neither follows from their conjunction; they are the commit-
897 ment this subsection names, and Section 9.2 lists the three observations that
898 would count against it.

899 Compatibility allows different descriptions to contribute different infor-
900 mation. It also permits uncertainty and conflicting judgments: apparent cup
901 position and judged cup position can be distinct quantities in a consistent
902 description of their discrepancy. Which quantities are shared must be de-
903 termined independently. Choosing them solely to secure agreement would
904 deprive the proposal of empirical force.

905 The minimal-self proposal concerns the perceiver’s position and condition
906 within the situation, such as the cup’s location relative to a body that can
907 reach it. The encoding could be implicit and need carry no autobiograph-
908 ical narrative. Its proposed necessity is philosophical: even an unreflective
909 episode has a perspective. Gluing by itself requires no self-encoding, and re-
910 constructability alone supplies no functional role. Altering the encoded hand
911 or viewpoint while holding the external scene fixed should change estimates
912 of the perceiver’s relation to the cup as the readout predicts. Whether that
913 represented perspective is experienced remains a further question.

914 **9.2 What a test of the full proposal must establish**

915 Both uniqueness results concern specified objects. They choose neither the
916 cover nor the encoding, and do not select between candidate subjects. In-

917 tegrated information theory treats exclusion explicitly (Tononi et al., 2016);
918 here selection requires independent evidence about neural organization. A
919 constant readout can contract and fix a predetermined state while discard-
920 ing its input. The family criterion in Section 5.2 tests the sensitivity that
921 this construction lacks, provided the relevant states and fidelity are declared
922 independently.

923 A test must separately measure coupling relative to diffusion, content
924 compatibility and self-reconstruction, alongside an independent endpoint for
925 experience. Meeting all three without a unified episode would challenge suffi-
926 ciency. A unified episode with incompatible descriptions would challenge the
927 unity identification; one lacking the specified self-encoding would challenge
928 the reflexive requirement. Awakening, dreaming and dissociative states offer
929 possible settings, but absence of report cannot by itself define absence of
930 experience.

931 The next physical requirement is a calibrated geometry–coupling relation
932 and resolved phase dynamics. The installation bound contributes a nec-
933 essary resource condition where its hardware representation applies; cortical
934 mode prices and field-mediated coupling require measurement. For the repre-
935 sentational connection, an informative experiment asks whether a controlled
936 change in coordination alters independently decoded content agreement, with
937 controls for sensory information and decoder quality. The constructed-data
938 estimator establishes feasibility for specified contents. Neural local descrip-
939 tions, shared overlap encoders and the self-encoding have yet to be identified.

940 The population-size dependence of the phase-to-content bound further limits
941 its quantitative use, in the form Section 4.2 reads off the sheet.

942 The separation results explain why these measurements must be inde-
943 pendent. A stationary distribution leaves several recovery mechanisms indis-
944 tinguishable. Equal phase order can accompany incompatible descriptions
945 or different task information. The tested plasticity rule preserves coherence
946 while losing its designated template. Each finding concerns the specified
947 model and observable; their role is to delimit the evidence needed for the
948 proposed connections.

949 **9.3 What the conditions constrain in a digital candi-** 950 **date**

951 The results of Section 2 hold for any candidate, and for a large language
952 model (LLM) running on graphics processing units (GPUs) four of them are
953 checkable rather than merely stateable. Those four are where this framework
954 bears on claims made for such systems. It does not exclude them, and no re-
955 sult here could; what it does is fix what a claim of unity or self-representation
956 commits its maker to measuring.

957 **Codes, not reports.** Reconstruction across a family is a counting con-
958 straint before it is anything else. If two macrostates the candidate is said
959 to represent satisfy $E(s) = E(t)$ while $d(s, t) > 2\varepsilon$, Eq. (10) rules out re-
960 constructing both within ε , whatever the readout. A system's reports are

961 readouts. Identical verbal reports across two conditions are therefore consis-
962 tent with identical internal codes and settle nothing about the states, while
963 the converse claim — that the system represents its own condition across a
964 declared family — asserts that its internal code separates that family, which
965 is a statement about activations under intervention and not about text. The
966 measurement follows from the statement: declare the family and the fidelity
967 first, then test whether the encoding region distinguishes its members.

968 **The region, not the readout.** Where the encoding is read as the state’s
969 restriction to a region of the machine, the demand falls on the region before
970 any readout exists. For a decoder-only transformer the causal mask makes
971 the network an instance of the update model of Section 5: the activation at
972 position t and layer ℓ reads positions at most t at layer $\ell - 1$ and nothing
973 else, so a region’s causal past is fixed by the architecture rather than chosen
974 by interpretation. The deadline bound of Section 5 therefore applies with
975 its communication graph given rather than assumed. A claim that a model
976 reports on its own processing consequently carries an obligation to say which
977 positions and layers the report’s causal past contains, and a report that
978 appears to track an intervention outside it indicates a confound or a different
979 region, not a faster channel.

980 **Agreement about what.** Unity requires the candidate’s local descriptions
981 to agree on independently identified shared quantities (Section 4). Common
982 timing, a shared residual stream and fluent output supply none of that, for

983 the reason Section 2 gives: coordination bounds content disagreement only
984 through encoders already related on overlaps, and which quantities two parts
985 of a model describe in common is a decoding question carrying the failure
986 modes of Section 7. The cover is a choice, and choosing it to secure agreement
987 empties the claim.

988 **What is not excluded.** Digital representations can satisfy compatibility,
989 so finite hardware is compatible with the gluing construction and none of
990 the above is a no-go for machines. The coupling-gated route E56–E78 is a
991 different matter: it requires a calibrated physical coupling, its noise regime
992 and a connection from coordination to content agreement, and an attention
993 matrix or a shared clock supplies none of these identifications. Comput-
994 ing a field model likewise leaves its realization in the computing substrate
995 to be established. Those physical obligations fall on the cortical proposal
996 identically.

997 The installed-energy condition of Section 3.2 is restricted in the same
998 way. It constrains a macroscopic candidate only through κ , which is hard-
999 ware data supplied by mode profiles and prices, and Supplemental Material
1000 reads the condition against a cortical metabolic budget to return the conver-
1001 sion factor it would require rather than a verdict. The hardware comparisons
1002 there have similarly restricted scope: fixed wiring can impose a correlation
1003 disadvantage, and point-supported kernels vanish under an atomless contin-
1004 uum measure, but neither comparison makes the representation conditions

1005 unattainable on GPUs. A universal exclusion would need a necessary condi-
1006 tion for consciousness and a proof that every relevant implementation violates
1007 it, and the present composition supplies neither.

1008 Whether experience requires a living substrate is the explicit subject of
1009 biological naturalism (Seth, 2026a,b). On that view the properties ground-
1010 ing experience are those of self-sustaining organisms, and simulating them
1011 does not secure them. The conditions stated here name no biological mech-
1012 anism, but computation described as such does not satisfy them either: a
1013 candidate must exhibit a physically realized coupling in a measured noise
1014 regime, agreement about independently identified shared quantities, and an
1015 internal reconstruction meeting a declared fidelity. Whether a realization
1016 of those conditions that is not alive would be accompanied by experience
1017 is the point at issue, and these results do not settle it. A related position
1018 locates the difference in the mode of computation rather than in life as such:
1019 biological systems compute across scales that do not separate, and along-
1020 side discrete operations carry continuous-valued dynamics belonging to their
1021 physical medium (Milinkovic & Aru, 2026). The coupling condition stated
1022 here is one measurable instance of that requirement, fixed to a physical field
1023 in a declared noise regime, and a description of such a field does not satisfy
1024 it.

1025 The neuroscientific case against *human-like* consciousness in text-only
1026 LLMs concerns embodied sensory interaction, thalamocortical organization
1027 and biological regulation linking perception and action to survival (Aru et al.,

2023). Their absence supports skepticism, conditional on their causal relevance to experience. Those mechanisms are not requirements of the abstract gluing and reconstruction definitions, and workspace-like representations reported in some LLMs (Gurnee et al., 2026) supply candidate organization to investigate. What this framework adds to that investigation is an order for the questions: declare the family of states and the fidelity, fix the encoding region and read its causal past off the architecture, and only then ask what the system reconstructs. A claim that survives those three steps is a claim about the machine; one that does not survive them was a claim about its output.

9.4 Explanatory scope

Integrated-information, global-workspace and predictive-processing accounts organize unity, access and modelling differently (Tononi et al., 2016; Baars, 1997; Dehaene et al., 2014; Friston, 2010). The present proposal contributes a candidate physical interaction and conditional relations between local content and self-encoding. Gluing and reconstruction permit multiple substrates; the electromagnetic commitment enters through physical calibration. The formalization checks implications under their hypotheses, and its witnesses establish realizability on declared models. Two things stand outside that result and are not supplied by it: the eight hypotheses are discharged for toy components rather than for cortex, and the identification of the glued state with unity and of its reconstruction with a minimal self is a further

1050 commitment that the eight do not contain.

1051 The explanatory question is whether the specified relations occur in neural
1052 activity and matter for experience. In the cup example, this means establish-
1053 ing how coordination helps maintain one represented situation and how the
1054 perceiver’s place within it contributes to the episode. The proposal identi-
1055 fies candidate correlates and tests of that identification. Why those relations
1056 would be accompanied by experience remains unexplained.

1057 **Data and code availability**

1058 The Lean development, simulation scripts, tests and saved numerical sum-
1059 maries are maintained in the GitHub repository `bobaseb/coupling_gated_`
1060 `cortical_coherence`. Publication simulation values are generated from
1061 saved summaries without rerunning production sweeps. The exploratory
1062 EEG analysis uses OpenNeuro dataset ds005620 (Bajwa et al., 2025); pre-
1063 processing and selection are described in Supplemental Material. The repos-
1064 itory contains the reproducible analysis and does not redistribute the raw
1065 recordings.

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1069 Use of AI tools

1070 AI assistance to this work was substantial and is not separable by task. Sev-
1071 eral coding assistants and chat interfaces were used in an interleaved way
1072 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
1073 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra; GPT-5.6 Terra,
1074 Luna and Sol), Claude Code (Claude Opus 5), Gemini Pro through its web in-
1075 terface, and Anthropic’s Fable and DeepSeek v4 Flash through OpenRouter,
1076 accessed with the Hermes CLI. Between them they contributed to the Lean
1077 development, the simulation and analysis code, and the drafting and revision
1078 of both documents. Because the tools were used in combination and in alter-
1079 nation, no section, proof, figure or result is attributable to any one of them,
1080 and none is claimed to be. The author is responsible for all content, including
1081 everything a tool produced. Machine checking establishes the stated Lean
1082 results under their hypotheses; it does not validate biological interpretations
1083 or replace scientific review of generated code and prose.

1084 References

- 1085 Pinotsis, D. A., & Miller, E. K. (2022). Beyond dimension reduction: Stable
1086 electric fields emerge from and allow representational drift. *NeuroImage*,
1087 253, 119058.
- 1088 Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated informa-

tion theory: from consciousness to its physical substrate. *Nature Reviews Neuroscience*, 17(7), 450-461.

Baars, B. J. (1997). In the Theater of Consciousness: The Workspace of the Mind. *Oxford University Press*.

Dehaene, S., Charles, L., King, J. R., & Marti, S. (2014). Toward a computational theory of conscious processing. *Current Opinion in Neurobiology*, 25, 76-84.

Friston, K. (2010). The free-energy principle: a unified brain theory?. *Nature Reviews Neuroscience*, 11(2), 127-138.

Milinkovic, B., & Aru, J. (2026). On biological and artificial consciousness: A case for biological computationalism. *Neuroscience & Biobehavioral Reviews*, 181, 106524. doi:10.1016/j.neubiorev.2025.106524.

Aru, J., Larkum, M. E., & Shine, J. M. (2023). The feasibility of artificial consciousness through the lens of neuroscience. *Trends in Neurosciences*, 46(12), 1008–1017. doi:10.1016/j.tins.2023.09.009.

Seth, A. K. (2026). Conscious artificial intelligence and biological naturalism. *Behavioral and Brain Sciences*, 49, e315. doi:10.1017/S0140525X25000032.

Seth, A. K. (2026). The stuff matters: Consciousness, computation, and biology. *Behavioral and Brain Sciences*, 49, e366. doi:10.1017/S0140525X26106906.

1109 Gurnee, W., et al. (2026). Verbalizable Representations Form a
1110 Global Workspace in Language Models. *Transformer Circuits Thread*.
1111 arXiv:2607.15495.

1112 Aguilera, M., Millidge, B., Tschantz, A., & Buckley, C. L. (2022). How par-
1113 ticular is the physics of the free energy principle?. *Physics of Life Reviews*,
1114 40, 24-50.

1115 Biehl, M., Pollock, F. A., & Kanai, R. (2021). A technical critique of some
1116 parts of the free energy principle. *Entropy*, 23(3), 293.

1117 Bruineberg, J., Dolega, K., Dewhurst, J., & Baltieri, M. (2022). The Em-
1118 peror’s New Markov Blankets. *Behavioral and Brain Sciences*, 45, e183.

1119 Pinotsis, D. A., & Miller, E. K. (2023). In vivo ephaptic coupling allows
1120 memory network formation. *Cerebral Cortex*, 33(17), 9877-9895.

1121 Pinotsis, D. A., Fridman, G., & Miller, E. K. (2023). Cytoelectric coupling:
1122 Electric fields sculpt neural activity and ”tune” the brain’s infrastructure.
1123 *Progress in Neurobiology*, 226, 102465.

1124 McFadden, J. (2002). The conscious electromagnetic information (cemi) field
1125 theory: the hard problem made easy?. *Journal of Consciousness Studies*,
1126 9(8), 45-60.

1127 Pockett, S. (2012). The electromagnetic field theory of consciousness: a
1128 testable hypothesis about the characteristics of conscious as opposed to
1129 non-conscious fields. *Journal of Consciousness Studies*, 19(11-12), 191-223.

- 1130 McFadden, J. (2020). Integrating information in the brain's EM field: the
1131 cemi field theory of consciousness. *Neuroscience of Consciousness*, 2020(1),
1132 niaa016.
- 1133 Fröhlich, F., & McCormick, D. A. (2010). Endogenous electric fields may
1134 guide neocortical network activity. *Neuron*, 67(1), 129-143.
- 1135 Anastassiou, C. A., & Koch, C. (2015). Ephaptic coupling to endogenous
1136 electric field activity: why bother?. *Current Opinion in Neurobiology*, 31,
1137 95-103.
- 1138 Pockett, S. (2002). Difficulties with the electromagnetic field theory of con-
1139 sciousness. *Journal of Consciousness Studies*, 9(4), 51-56.
- 1140 Vöröslakos, M., et al. (2018). Direct effects of transcranial electric stimulation
1141 on brain circuits in rats and humans. *Nature Communications*, 9, 483.
- 1142 Verkhratsky, A., & Nedergaard, M. (2018). Physiology of Astroglia. *Physio-*
1143 *logical Reviews*, 98(1), 239-389.
- 1144 Mountcastle, V. B. (1997). The columnar organization of the neocortex.
1145 *Brain*, 120(4), 701-722. doi:10.1093/brain/120.4.701.
- 1146 Horton, J. C., & Adams, D. L. (2005). The cortical column: a structure
1147 without a function. *Philosophical Transactions of the Royal Society B*,
1148 360(1456), 837-862. doi:10.1098/rstb.2005.1623.

1149 Rakic, P. (2008). Confusing cortical columns. *Proceedings of the National*
1150 *Academy of Sciences*, 105(34), 12099-12100. doi:10.1073/pnas.0807271105.

1151 Bastos, A. M., Donoghue, J. A., Brincat, S. L., Mahnke, M., Yanar, J., Cor-
1152 rea, J., ... & Miller, E. K. (2021). Neural effects of propofol-induced uncon-
1153 sciousness and its reversal using thalamic stimulation. *eLife*, 10, e60824.

1154 Bajwa, I. J., Nilsen, A. S., Skukies, R., Aamodt, A., Ernst, G., Storm, J.
1155 F., & Juel, B. E. (2025). A repeated awakening study exploring the capac-
1156 ity of complexity measures to capture dreaming during propofol sedation.
1157 *Scientific Reports*, 15, 32746.

1158 Xie, L., Kang, H., Xu, Q., Chen, M. J., Liao, Y., Thiyagarajan, M., ... &
1159 Nedergaard, M. (2013). Sleep drives metabolite clearance from the adult
1160 brain. *Science*, 342(6156), 373-377.

1161 Syková, E., & Nicholson, C. (2008). Diffusion in brain extracellular space.
1162 *Physiological Reviews*, 88(4), 1277-1340.

1163 Barbour, B. (2017). Analysis of claims that the brain extracellular impedance
1164 is high and non-resistive. *Biophysical Journal*, 113(7), 1636-1638.

1165 Norton, J. D. (2005). Eaters of the lotus: Landauer's principle and the return
1166 of Maxwell's demon. *Studies in History and Philosophy of Science Part B:*
1167 *Studies in History and Philosophy of Modern Physics*, 36(2), 375-411.

1168 Landauer, R. (1961). Irreversibility and heat generation in the computing
1169 process. *IBM Journal of Research and Development*, 5(3), 183-191.

1170 Kibble, T. W. B. (1976). Topology of cosmic domains and strings. *Journal*
1171 *of Physics A: Mathematical and General*, 9(8), 1387-1398.

1172 Sakaguchi, H. (1988). Cooperative phenomena in coupled oscillator systems
1173 under external fields. *Progress of Theoretical Physics*, 79(1), 39-46.

1174 Still, S., Sivak, D. A., Bell, A. J., & Crooks, G. E. (2012). Thermodynamics
1175 of prediction. *Physical Review Letters*, 109(12), 120604.

1176 Kawai, R., Parrondo, J. M. R., & Van den Broeck, C. (2007). Dissipation:
1177 The phase-space perspective. *Physical Review Letters*, 98(8), 080602.

1178 Horowitz, J. M., & Esposito, M. (2014). Thermodynamics with continuous
1179 information flow. *Physical Review X*, 4, 031015. doi:10.1103/PhysRevX.4.
1180 031015.

1181 Ito, S., & Sagawa, T. (2013). Information thermodynamics on
1182 causal networks. *Physical Review Letters*, 111, 180603. doi:10.1103/
1183 PhysRevLett.111.180603.

1184 Corberi, F., dello Russo, S., Messuti, G., Scarpetta, S., & Smaldone, L.
1185 (2026). Thermodynamic learning. *arXiv preprint*. arXiv:2609.04732.

1186 Banach, S. (1922). Sur les opérations dans les ensembles abstraits et leur ap-
1187 plication aux équations intégrales. *Fundamenta Mathematicae*, 3(1), 133-
1188 181.

- 1189 Shenker, O. R. (2000). Logic and entropy. *PhilSci-Archive preprint*. philsci-
1190 archive.pitt.edu/115.
- 1191 Rosas, F. E., Mediano, P. A. M., Luppi, A. I., Boyd, A. B., Berta, M., Jensen,
1192 H. J., Gastpar, M., & Carroll, S. M. (2026). Four faces of information in
1193 natural systems. *Nature Reviews Physics*. doi:10.1038/s42254-026-00971-4.
- 1194 Sznitman, A.-S. (1991). Topics in propagation of chaos. In *École d'Été de*
1195 *Probabilités de Saint-Flour XIX—1989*, Lecture Notes in Mathematics,
1196 vol. 1464, pp. 165–251. Springer. doi:10.1007/BFb0085169.
- 1197 Berglund, N., & Gentz, B. (2002). Pathwise description of dynamic pitchfork
1198 bifurcations with additive noise. *Probability Theory and Related Fields*,
1199 122, 341–388. doi:10.1007/s004400100174.
- 1200 Abramsky, S., & Brandenburger, A. (2011). The sheaf-theoretic structure
1201 of non-locality and contextuality. *New Journal of Physics*, 13, 113036.
1202 doi:10.1088/1367-2630/13/11/113036.
- 1203 Townsend, R. G., Solomon, S. S., Chen, S. C., Pietersen, A. N. J., Martin,
1204 P. R., Solomon, S. G., & Gong, P. (2015). Emergence of complex wave
1205 patterns in primate cerebral cortex. *The Journal of Neuroscience*, 35(11),
1206 4657-4662. doi:10.1523/JNEUROSCI.4509-14.2015.
- 1207 Xu, Y., Long, X., Feng, J., & Gong, P. (2023). Interacting spiral wave pat-
1208 terns underlie complex brain dynamics and are related to cognitive pro-

- 1209 cessing. *Nature Human Behaviour*, 7(7), 1196-1215. doi:10.1038/s41562-
1210 023-01626-5.
- 1211 Nicoletti, G., Di Terlizzi, I., & Busiello, D. M. (2026). Balancing information
1212 and dissipation with partially observed fluctuating signals. *Physical Review*
1213 *Letters*, 137(12), 127101. doi:10.1103/8gx2-rbns.
- 1214 Kiyooka, D., Oomoto, I., Kitazono, J., Saito, Y., Kobayashi, M., Matsub-
1215 ara, C., Kobayashi, K., Murayama, M., & Oizumi, M. (2026). Single-
1216 cell resolution functional networks during unconsciousness are segre-
1217 gated into spatially intermixed modules. *Cell Reports*, 45(2), 116902.
1218 doi:10.1016/j.celrep.2025.116902.
- 1219 Oomoto, I., Kiyooka, D., Oizumi, M., & Murayama, M. (2026). Multi-area
1220 single-cell calcium imaging dataset of the mouse cortex across wakefulness,
1221 sleep, and anesthesia. *Scientific Data*. doi:10.1038/s41597-026-08202-2.
- 1222 Attwell, D., & Laughlin, S. B. (2001). An energy budget for signaling in the
1223 grey matter of the brain. *Journal of Cerebral Blood Flow & Metabolism*,
1224 21(10), 1133-1145. doi:10.1097/00004647-200110000-00001.
- 1225 Harris, J. J., Jolivet, R., & Attwell, D. (2012). Synaptic energy use and
1226 supply. *Neuron*, 75(5), 762-777. doi:10.1016/j.neuron.2012.08.019.
- 1227 Nicholls, D. G., & Ferguson, S. J. (2013). *Bioenergetics 4*. Academic Press.